

Gaussian tails for a covariant linearized flow; Hamiltonian truncation remains open

Later corrections: the instability note withdraws exact sharpness; the lattice note separates one lattice step from fixed-scale continuum bounds; the operator-inequality note supersedes the fixed-range competition in the transfer note.

Correction (2026-10-02). The original title, lead and Corollary 3 identified a Gaussian Jacobian-kernel tail with a relative error of the conjugated Hamiltonian and claimed it was small **exactly** on small fields. Those operator and only-if claims are **withdrawn**. The Nielsen–Olesen example grows at gB , not $2gB$, and supplies no spatial tail lower bound. Proposition 1 now retains the variation of Lüscher’s gauge term; the simple covariant heat equation is a separately specified background-covariant linear evolution, not the unrestricted Jacobian of that gauge-modified flow. Proposition 2 uses the full curvature block-operator norm and time-dependent propagators. Sections 3–4 give an absolute kernel-tail bound, with its prefactor, rather than an operator relative bound. The lattice-step claim at one flow radius, the scale substitution and the reversed free-smoothing comparison are corrected below. The claim that Hamiltonian methods have no access to configuration probabilities is also withdrawn.

For a smooth compact-Lie-algebra connection on $\mathbb{R}^3$, consider specifically the linear evolution

$$\partial_s u_\mu = D(s)^2 u_\mu + 2[G_{\mu\nu}(s), u_\nu], \quad D_\mu = \partial_\mu + [B_\mu, \cdot].$$

It is obtained by adding a background-covariant gauge term to the ungauged linearization. Its propagator $J(t, 0)$ has a Gaussian kernel majorant under the regularity and bounded-curvature hypotheses of Section 2. Define the pointwise curvature operator on vector fields by

$$(\mathcal{C}(s, x)u)_\mu = \sum_{\nu=1}^3 [G_{\mu\nu}(s, x), u_\nu], \quad \Gamma(s) = \sup_x \|\mathcal{C}(s, x)\|_{\text{op}}, \quad A_t = 2 \int_0^t \Gamma(s) ds.$$

Then

$$\|J(t, 0; x, y)\|_{\text{op}} \leq e^{A_t} \frac{e^{-|x-y|^2/(4t)}}{(4\pi t)^{3/2}}.$$

For $R = \kappa\sqrt{8t}$ the absolute discarded kernel mass is bounded by

$$e^{A_t} \left[\text{erfc}(\sqrt{2}\kappa) + \frac{2\sqrt{2}\kappa}{\sqrt{\pi}} e^{-2\kappa^2} \right].$$

This is a sufficient range/curvature budget for this linear evolution. It is not a necessary small-field condition, a bound on the actual Jacobian in every gauge, or a relative form estimate for a conjugated kinetic operator. No continuum construction or mass gap follows.

Sharpness corrected (2026-10-02). In the constant chromomagnetic model of the instability note, §2 (full-read), $\Gamma = b = gB$, the curvature term has eigenvalue $2b$, but the covariant Laplacian contributes $-b - k_\parallel^2$ on that mode. The full growth rate is $b - k_\parallel^2$. This does not saturate e^{2bt} or prove a lower bound on the discarded spatial tail.

Lattice scope corrected (2026-10-02). Even granting the earlier schematic normalization $a^2\Gamma \leq \pi$, at $t = a^2/2$ and $R = \sqrt{8t}$ the exponent budget is $\pi - 2 > 0$, not small. Compactness gives a finite coefficient budget, not accuracy at one flow

radius. Increasing κ controls the continuum majorant for any fixed finite budget; merely taking $\kappa > 1$ is not a specified accuracy guarantee. The lattice note, §§1–4 (full-read), discusses bounded coefficients and longer lattice range. A plaquette-angle bound is not by itself a proof of the above continuum block-norm bound or a lattice Hamiltonian relative bound; those require their own normalization and operator estimates.

1. The linearized flow and its gauge convention

Take Lüscher’s gauge-modified continuum flow

$$\partial_s B_\mu = D_\nu G_{\nu\mu} + D_\mu q, \quad q = \partial_\nu B_\nu.$$

The definition is from arXiv:1006.4518v3, equations (1.1)–(1.2) and (2.2), passage level via the local companion. Repeated spatial indices are summed. Write $u = \delta B$.

Proposition 1 (exact variation and auxiliary gauge choice). The Jacobian of this gauge-modified flow obeys

$$\partial_s u_\mu = D^2 u_\mu + 2[G_{\mu\nu}, u_\nu] + D_\mu(\partial_\nu u_\nu - D_\nu u_\nu) + [u_\mu, q].$$

The ungauged flow instead has linearization

$$\partial_s u_\mu = D^2 u_\mu - D_\mu D_\nu u_\nu + 2[G_{\mu\nu}, u_\nu].$$

Adding the **linear background-covariant gauge term** $D_\mu D_\nu u_\nu$ to the latter defines the auxiliary evolution used in Sections 2–4. Identifying it with another flow Jacobian requires an additional gauge comparison, not supplied here.

Proof. Since $\delta D_\nu = [u_\nu, \cdot]$ and $\delta G_{\nu\mu} = D_\nu u_\mu - D_\mu u_\nu$,

$$\delta(D_\nu G_{\nu\mu}) = D^2 u_\mu - D_\nu D_\mu u_\nu + [u_\nu, G_{\nu\mu}] = D^2 u_\mu - D_\mu D_\nu u_\nu + 2[G_{\mu\nu}, u_\nu].$$

Here $[D_\nu, D_\mu]u_\nu = [G_{\nu\mu}, u_\nu]$. The gauge term has the exact variation

$$\delta(D_\mu q) = D_\mu \partial_\nu u_\nu + [u_\mu, q],$$

not $D_\mu D_\nu u_\nu$. Adding these formulas proves the first identity; omitting the gauge term or adding the specified linear one proves the other assertions. $\square$

Thus the original cancellation “to the same order” was unjustified: the omitted terms are themselves linear in u . The propositions below are about J , not about the full $D\Phi_t$ in Lüscher’s convention. At $B = 0$ both equations reduce to the free heat equation. At a stationary background, background-transverse modes of the ungauged linearization also agree with the auxiliary generator; no general preservation of that transverse condition is asserted.

2. The Gaussian bound

Equip the compact Lie algebra with an invariant positive inner product, and its vector fields with $|u|^2 = \sum_\mu |u_\mu|^2$. The connection is metric-compatible because $\text{ad}(B_\mu)$ is skew-adjoint. Assume $B(s, x)$ is smooth on $[0, t] \times \mathbb{R}^3$, with bounded coefficients and spatial derivatives sufficient for the covariant parabolic propagators and kernels below, and assume $\Gamma \in L^1([0, t])$. These are hypotheses, not estimates derived from action monotonicity.

Proposition 2. Under these assumptions the auxiliary propagator satisfies

$$\|J(t, 0; x, y)\|_{\text{op}} \leq e^{A_t} K_t^{\text{free}}(x - y), \quad K_t^{\text{free}}(z) = (4\pi t)^{-3/2} e^{-|z|^2/(4t)}.$$

In particular, if $\Gamma(s) \leq \Gamma_*$ then $A_t \leq 2t\Gamma_*$.

Proof. First consider $\partial_s v = D(s)^2 v$. Metric compatibility gives

$$(\partial_s - \Delta)|v|^2 = -2 \sum_{\alpha} |D_{\alpha} v|^2.$$

Apply the chain rule to $h_{\epsilon} = (|v|^2 + \epsilon^2)^{1/2}$ and use $|\partial_{\alpha}|v|^2| \leq 2|v||D_{\alpha} v|$. It follows that $(\partial_s - \Delta)h_{\epsilon} \leq 0$. Scalar comparison followed by $\epsilon \downarrow 0$ yields, for the time-dependent covariant propagator $U(t, r)$,

$$|U(t, r)f|(x) \leq (e^{(t-r)\Delta}|f|)(x).$$

With the assumed kernels this also gives $\|U(t, r; x, y)\|_{\text{op}} \leq K_{t-r}^{\text{free}}(x - y)$, by testing localized initial data. This proves the needed Kato domination here; no bibliographic sharpness claim is used.

Set $M(s) = 2\mathcal{C}(s)$. Duhamel uses the two-time propagator,

$$J(t, 0) = U(t, 0) + \int_0^t U(t, s)M(s)J(s, 0) ds.$$

In the term with n insertions, each multiplication factor has norm at most $2\Gamma(s_i)$ and scalar heat kernels convolve to $K_t^{\text{free}}(x - y)$. The ordered time integral is bounded by $A_t^n/n!$. Summing this absolutely bounded series proves the result. $\square$

Norm convention. If instead $g_* = \sup_{s, x, \mu, \nu} \|\text{ad}(G_{\mu\nu}(s, x))\|_{\text{op}}$, then $\Gamma_* \leq 2g_*$ in three dimensions, or $\Gamma_* \leq (d-1)g_*$ in d dimensions. Indeed

$$|(\mathcal{C}u)_{\mu}| \leq g_* \sum_{\nu \neq \mu} |u_{\nu}|,$$

and Cauchy–Schwarz, followed by summation over μ , gives $|\mathcal{C}u|^2 \leq (d-1)^2 g_*^2 |u|^2$. The former unspecified “adjoint operator norm” cannot silently omit this index factor.

At $B = 0$, $J = K_t^{\text{free}}I$ and the pointwise bound is attained. This free-case equality does not establish optimality of the curvature exponent for nonzero backgrounds.

3. Absolute kernel tails, not Hamiltonian relative errors

Define J_R by cutting its kernel to $|x - y| \leq R$, and set

$$T_R = J - J_R, \quad \delta(R, t) = e^{A_t} \int_{|z| > R} K_t^{\text{free}}(z) d^3 z.$$

Proposition 2 bounds both row and column integrals of $\|T_R(x, y)\|_{\text{op}}$ by $\delta(R, t)$. Consequently the Schur test (or the scalar convolution majorant) gives the **absolute** bound

$$\|T_R\|_{L^2 \rightarrow L^2} \leq \delta(R, t).$$

It is not normalized by $\|J\|$ or by a kinetic quadratic form.

Corollary 3 (kernel-tail budget). Put $q = R/(2\sqrt{t})$. Then

$$\delta(R, t) = e^{A_t} \left[\text{erfc}(q) + \frac{2q}{\sqrt{\pi}} e^{-q^2} \right] \leq e^{A_t - q^2} \left(1 + \frac{2q}{\sqrt{\pi}} \right).$$

Thus at $R = \kappa\sqrt{8t}$ a sufficient condition for $\|J - J_R\|_{2 \rightarrow 2} \leq \epsilon$ is

$$2\kappa^2 \geq A_t + \log \left(1 + \frac{2\sqrt{2}\kappa}{\sqrt{\pi}} \right) + \log(1/\epsilon), \quad 0 < \epsilon < 1.$$

For every finite A_t this is achieved at sufficiently large κ . It is a sufficient condition only.

Proof. Radial integration and $r = 2\sqrt{t}v$ give

$$\int_{|z|>R} K_t^{\text{free}}(z) d^3z = \frac{4}{\sqrt{\pi}} \int_q^\infty v^2 e^{-v^2} dv = \text{erfc}(q) + \frac{2q}{\sqrt{\pi}} e^{-q^2}.$$

For $q \geq 0$, $\text{erfc}(q) \leq e^{-q^2}$: their difference is zero at zero, has derivative $2e^{-q^2}(1/\sqrt{\pi} - q)$, and tends to zero at infinity. The stated inequality and logarithmic budget follow. $\square$

At $\kappa = 1$, even the exact free discarded mass is $\text{erfc}(\sqrt{2}) + (2\sqrt{2}/\sqrt{\pi})e^{-2}$. This is a fixed positive number, not a parameter tending to zero on small fields. A growing upper bound for A_t implies neither a large actual tail nor a lower bound on a truncation error.

The Hamiltonian conversion remains open. The kinetic metric in the flow-conjugation note, §§1–3 (full-read), involves quadratic Jacobian combinations and inverse metrics; the transformed operator also has density derivatives and lower-order terms. Already for bounded operators F and F_R , with $E = F - F_R$,

$$\|F^*F - F_R^*F_R\| \leq 2\|F\| \|E\| + \|E\|^2.$$

This follows by expanding $F_R = F - E$. It is an absolute estimate; a relative comparison additionally needs a positive lower bound on the relevant metric or a bound on its inverse. The free continuum heat multiplier $e^{-t|k|^2}$ has no bounded inverse on L^2 . The finite lattice has an invertible flow Jacobian, but uniform inverse, configuration-derivative and relative form estimates still need proof. The auxiliary J above is not identified with that lattice Jacobian. Thus the original Hamiltonian interpretation of δ remains a **heuristic motivation**, not a proved relative truncation theorem.

4. The sufficient small-field budget and its limits

Write $\ell = \sqrt{8t}$. If the block norm satisfies

$$\Gamma(s) \leq \frac{\eta}{\ell^2} \quad (0 \leq s \leq t),$$

then $A_t \leq \eta/4$ and Corollary 3 yields

$$\|J - J_R\|_{2 \rightarrow 2} \leq \left(1 + \frac{2\sqrt{2}\kappa}{\sqrt{\pi}}\right) \exp\left[\frac{\eta}{4} - 2\kappa^2\right].$$

The original 2η growth exponent in this substitution was incorrect. In component norm, the hypothesis $g_* \leq \eta/\ell^2$ gives instead $A_t \leq \eta/2$ in three dimensions by the index bound of Section 2.

regime or question	supported conclusion
bounded curvature budget A_t , sufficiently long range	absolute auxiliary kernel tail is below a prescribed tolerance
small η at one fixed flow radius	bounded tail, with no arbitrarily small accuracy implied
large A_t at fixed range	this upper bound deteriorates; actual tail size remains undetermined
larger range at finite A_t	the kernel majorant can be made small
conjugated Hamiltonian or iterated RG	relative form and iteration estimates remain open

This budget motivates comparison with small/large-field splitting in constructive renormalization, but neither forces that split nor controls all steps of the flow-conjugation route. The comparison to Balaban’s series (Commun. Math. Phys. 122 (1989) 355, metadata level) is context, not a derivation of its hypotheses. No unproved decay of a “typical” interacting curvature is used here.

5. What action monotonicity controls

For smooth finite-action solutions of the ungauged gradient flow, with decay allowing integration by parts and the gradient normalization,

$$\frac{d}{ds}S(B_s) = -\|\nabla S(B_s)\|_2^2 \leq 0.$$

The added gauge term is an infinitesimal gauge transformation and does not change this identity under the same boundary assumptions. Since S is proportional to $\int |G|^2$, it gives

$$\|G(t)\|_2^2 \leq \|G(0)\|_2^2.$$

It supplies no L^∞ bound by itself, and Proposition 2 requires control over the **whole flow interval**, not just its endpoint.

Free or abelian smoothing, corrected. Around the zero connection J is the heat semi-group. In an abelian theory the curvature itself obeys the heat equation, so Cauchy–Schwarz and the Gaussian convolution identity give

$$\|e^{t\Delta}f\|_\infty \leq \|K_t^{\text{free}}\|_2 \|f\|_2 = (8\pi t)^{-3/4} \|f\|_2.$$

For abelian initial curvature this bounds $\|G(t)\|_\infty$ by $(8\pi t)^{-3/4}\|G(0)\|_2$ in the Euclidean component norm. For fixed nonzero initial norm its ratio to $1/t$ is proportional to $t^{1/4}\|G(0)\|_2$: it tends to zero for small t and grows for large t , the reverse of the former remark. Integrating this estimate over $0 < s < t$ also gives a finite budget proportional to $t^{1/4}\|G(0)\|_2$, using the finite-dimensional conversion to the block norm. This is an abelian estimate, not an interacting theorem. The coefficient $(4\pi t)^{-3/4}$ formerly displayed is a looser valid bound; the expression above is the exact heat-kernel L^2 norm.

Monotonicity alone does not supply the missing norm. A fixed L^2 curvature norm permits concentration in small regions with arbitrarily large height. For example rescale a smooth compactly supported abelian connection by $B_\lambda(x) = \lambda^{1/2}B(\lambda x)$; its curvature is $G_\lambda(x) = \lambda^{3/2}G(\lambda x)$, with constant L^2 norm and unbounded supremum as $\lambda \rightarrow \infty$. This disproves a pointwise estimate from the action bound alone, not parabolic estimates using additional evolution information.

The Hamiltonian formulation also has configuration probabilities, for example its ground-state density; see the transfer note, §§1–2 (full-read). Such measure estimates do not automatically give the signed operator comparisons needed for spectral stability. The distinction emphasized in the operator-inequality note, §3 (full-read), is relevant here; this note does not assume its claimed Hamiltonian truncation estimate or conclude that every Hamiltonian route fails.

6. Consequence for STATE

The surviving result is a Gaussian majorant and explicit absolute kernel-tail budget for the specified background-covariant linear parabolic evolution, with a full vector curvature norm and a flow-interval bound. Small fields give a sufficient budget at suitable range; neither necessity nor sharpness is proved. The identification with Lüscher’s unrestricted flow Jacobian and the conversion to a relative Hamiltonian truncation error remain open, as do lattice normalization, inverse-metric and density-derivative control and iteration. Single-step compact lattice bounds must be distinguished from fixed-scale continuum estimates. This correction supplies no continuum $SU(3)$ construction or mass gap. STATE and the catalogs are left to the caller, as requested.